

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Some relevant points made from any of the description areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit.						
	(b)	(i)		Substitution: Momentum change = $10\,000 \times (25 - 0)(1)$ = $250\,000$ [kg m/s] (1)	1	1		2	2	
		(ii)		Substitution: $F = \frac{\Delta p}{t} = \frac{250\,000(\text{ecf})}{50}$ (1) = $5\,000$ [N](1) Alternative: $a = 25 \div 50 = 0.5$ [m/s ²] (1) $F = ma = 0.5 \text{ ecf} \times 10\,000 = 5\,000$ [N] (1) [NB. ecf only on a <u>calculation</u> of acceleration, not on a change in velocity]	1	1		2	2	
		(iii)		Substitution: Distance = 25×80 (1) [No tolerance] = $2\,000$ [m] (1)	1	1		2	2	
		(iv)		The line / it would be lower on the graph [than BC] (1) The length [of the line] would be greater [than 80 s](1)			2	2	1	
	(c)			Bigger mass / more inertia [not: heavier] (1) [so the same force] would produce a <u>smaller</u> [not: slower] acceleration (1)	2			2		
				Question 7 total	8	6	2	16	10	0